Value investing: buy-and-hold results

S&P MidCap 400, data as of 2026-10-31

Abstract

Each January and July since 2016, a rule-based value score picks 5 S&P MidCap 400 companies, and 100 USD is invested in them, weighted by score and never sold. Investors who started in 2016, 2020, 2023 and 2025 (an earlier start buys from 2016) are followed to 30 October 2026, each against the same money in the iShares Core S&P Mid-Cap fund (IJH), dividends reinvested, net of its 0.05% yearly fee, and in Berkshire Hathaway. The reference is the investor who started in 2020: the data hold only today's index members, and by the index's list of changes, which starts in January 2012, 293 companies have left the index since then against 468 since 2016, so fewer failed companies are missing. That investor's 1,400 USD is worth 2,400 USD, against 2,073 USD in the benchmark and 2,415 USD in Berkshire Hathaway; a second wallet that also sells holdings whose score has fallen is worth 2,383 USD. The report closes with the next picks: UFP Industries (UFPI), Ingredion (INGR), Simpson Manufacturing (SSD), Harley-Davidson (HOG) and Reliance, Inc. (RS). The result rests on few decisions and few companies, with 25% of the value in one company; it is a research result, not evidence of a lasting edge.

1 Method

Company fundamentals are the values first reported in each annual report (10-K) to the SEC, used from the day they were filed. When a company reports an item under more than one label, the label it used most recently by the date is taken, so later filings never decide it (Appendix D). Prices and dividends are from Yahoo Finance; returns include reinvested dividends and are in US dollars. On each date, only companies already in the S&P MidCap 400 are considered, outside the Financials and Real Estate sectors.

Sixteen rules from Graham, Buffett and a net income growth test grade every company. For each rule, the companies are ranked by the rule's value and placed in six equal groups, from 5 points for the best sixth to 0 for the worst; each rule's points are multiplied by its weight, one less the share of companies that pass the rule's threshold, so rules few companies pass count more. The score is the sum. On the first trading day of January and of July, the 5 highest-scoring companies are bought for 100 USD, each in proportion to its score.

Three limits apply to every number in this report. Companies that left the index are missing from the data, which flatters the results. The score's design and every setting of this report (the rules' weighting, the five picks, the decision months, the sell threshold, the start years) were chosen while looking at the whole period from 2016, so no period here is out of sample: the results describe how these rules would have done, not evidence that they will do so. There are no trading costs or taxes.

2 Picks per half-year

Table 1: Purchases per half-year and their value today (USD).

Bought	Picks and weights	Wallet	S&P MidCap 400	Berkshire
2016-01	CMC 23%, AVT 21%, TKR 19%, RRX 19%, PPC 19%	374	313	415
2016-07	CMC 21%, GNTX 21%, AVT 20%, SNX 20%, BC 18%	334	285	377
2017-01	GNTX 21%, AVT 21%, CMC 20%, SNX 19%, BC 19%	256	254	331
2017-07	GAP 22%, GNTX 21%, AVT 20%, CMC 19%, SNX 18%	263	240	316
2018-01	AVT 23%, GNTX 21%, SYNA 20%, ENS 18%, GAP 18%	204	218	275
2018-07	GNTX 23%, AVT 21%, RS 19%, ARW 19%, GAP 18%	258	212	289
2019-01	GNTX 23%, MKSI 20%, OSK 19%, MMS 19%, CMC 19%	245	248	267
2019-07	GNTX 22%, MKSI 21%, CMC 20%, ARW 19%, INGR 19%	236	209	253
2020-01	AVT 20%, ARW 20%, MKSI 20%, INGR 20%, GAP 20%	194	196	238
2020-07	RS 21%, AVT 21%, AYI 20%, INGR 19%, CMC 18%	354	227	305
2021-01	RS 22%, INGR 21%, CMC 20%, MMS 19%, LSTR 18%	225	176	237
2021-07	LOPE 21%, INGR 20%, RS 20%, CMC 20%, LPX 19%	190	146	194
2022-01	CMC 22%, LOPE 20%, INGR 20%, RS 19%, AVT 18%	198	138	180
2022-07	RS 21%, LOPE 21%, ITT 20%, CMC 20%, MKSI 19%	230	170	195
2023-01	MKSI 21%, RS 21%, IPGP 20%, AVT 19%, CMC 19%	185	160	175
2023-07	UFPI 23%, RS 22%, CMC 19%, AVT 18%, WLK 18%	127	146	159
2024-01	UFPI 21%, CMC 21%, RS 21%, AVT 19%, IPGP 18%	124	137	150
2024-07	RS 21%, UFPI 21%, IPGP 20%, CMC 20%, MSM 19%	123	130	134
2025-01	MLI 21%, UFPI 20%, RS 20%, CMC 20%, TEX 18%	129	121	120
2025-07	MLI 22%, UFPI 21%, CMC 20%, RS 19%, WLK 18%	114	119	111
2026-01	UFPI 22%, MLI 20%, RS 20%, WLK 19%, INGR 18%	104	110	109
2026-07	UFPI 21%, HOG 20%, INGR 20%, MLI 20%, SSD 19%	103	96	109

Table 1 lists the companies bought each January and July, with their weights, and what each 100 USD is worth on 30 October 2026, next to the same money in the S&P MidCap 400 and in Berkshire Hathaway. Figure 1 shows the same values. Most of the money went to Materials, Information Technology and Industrials; the best single purchase is CMC in January 2016, now worth 6 times its cost. The purchases of January 2018, January 2019, January 2020, July 2023, January 2024, July 2024, July 2025 and January 2026 are worth less today than the same money in the S&P MidCap 400. Appendix A lists every purchase of the wallet with sells of the investor who started in 2020, with its price at purchase and today, what later sales took from it, what it keeps today and its total return with dividends.

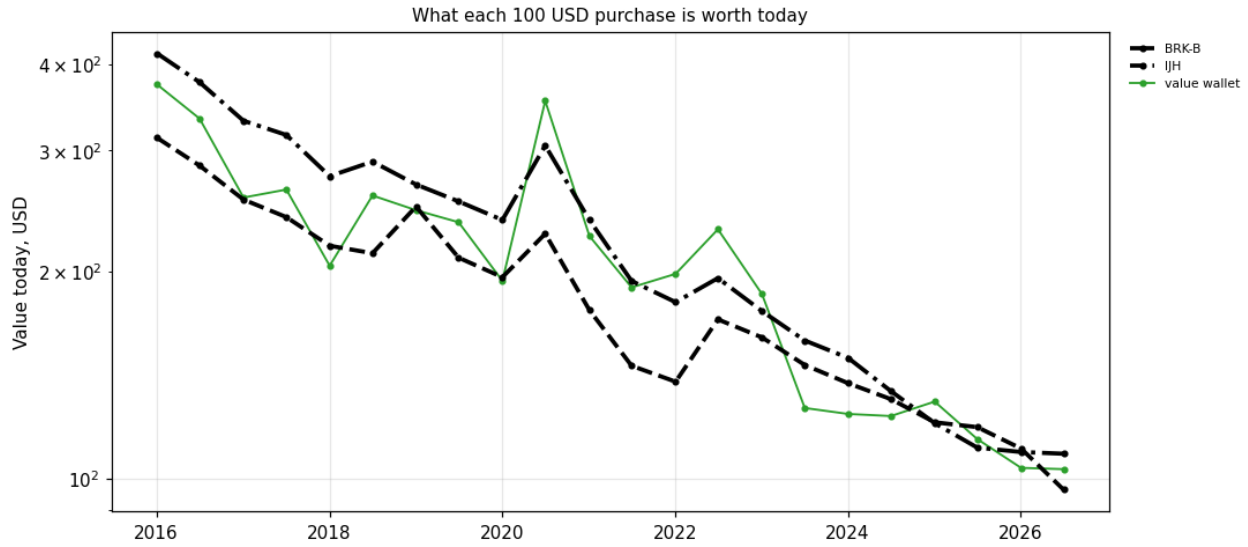


Figure 1: What each purchase is worth today, by date of purchase, on a logarithmic scale.

3 Value over time

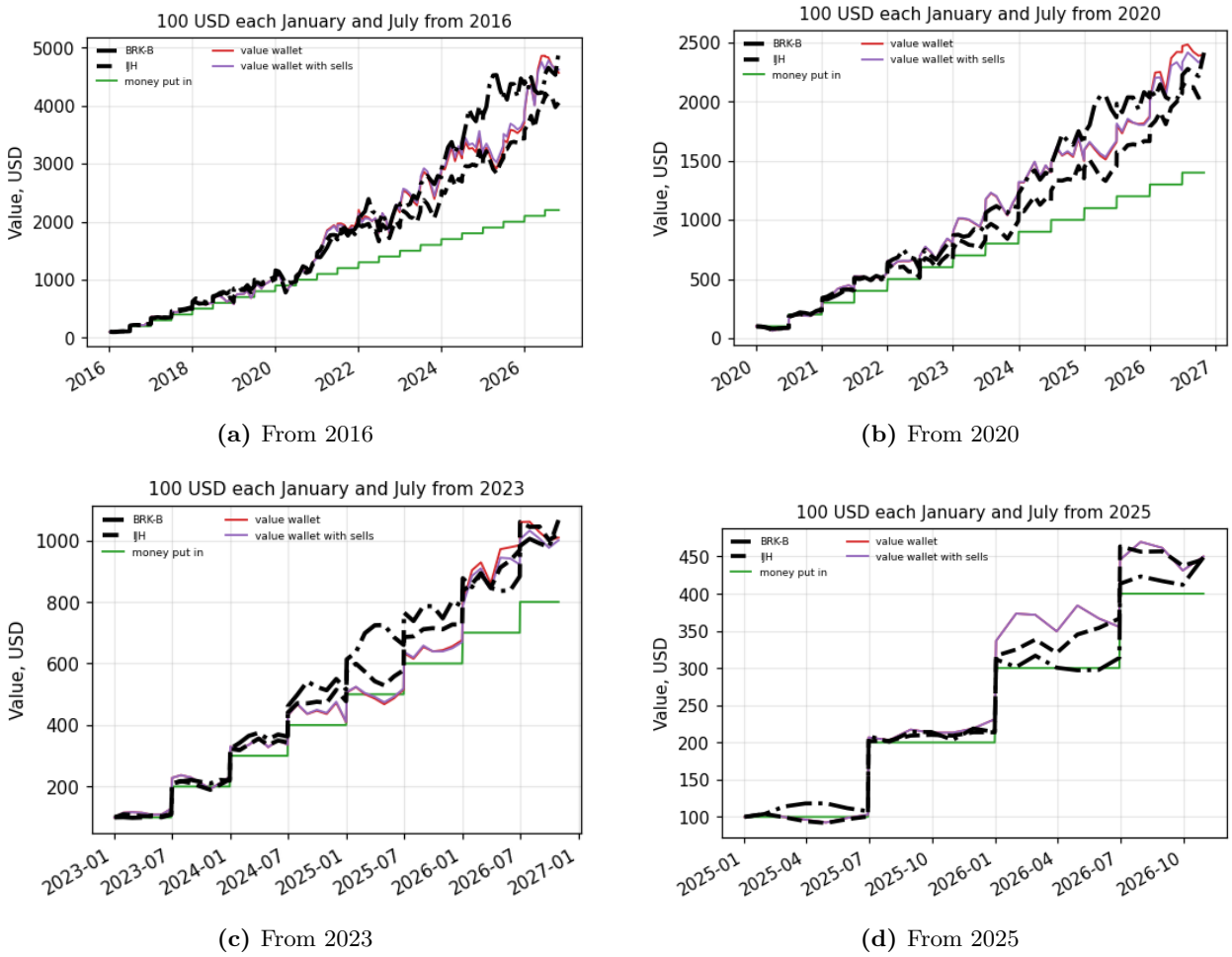
Table 2: Money put in and value today per start year (USD), and the money-weighted return a year.

Start	Holder	Put in	Value	Multiple	A year
2016	Value wallet	2,200	4,569	2.08	12.6%
	Value wallet with sells	2,200	4,623	2.10	12.8%
	S&P MidCap 400	2,200	4,053	1.84	10.6%
	Berkshire Hathaway	2,200	4,938	2.24	13.9%
2020	Value wallet	1,400	2,400	1.71	15.0%
	Value wallet with sells	1,400	2,383	1.70	14.8%
	S&P MidCap 400	1,400	2,073	1.48	10.9%
	Berkshire Hathaway	1,400	2,415	1.73	15.2%
2023	Value wallet	800	1,009	1.26	11.4%
	Value wallet with sells	800	1,000	1.25	11.0%
	S&P MidCap 400	800	1,020	1.28	12.0%
	Berkshire Hathaway	800	1,066	1.33	14.2%
2025	Value wallet	400	450	1.12	11.3%
	Value wallet with sells	400	450	1.12	11.4%
	S&P MidCap 400	400	446	1.11	10.5%
	Berkshire Hathaway	400	449	1.12	11.1%

Table 2 and Figure 2 follow the four investors. Each puts in 100 USD every January and July from the start year; the value wallet never sells. The return a year is money-weighted: the constant yearly rate that turns the payments into today's value.

The investor who started in 2016 earns 12.6% a year against 10.6% in the S&P MidCap 400 and 13.9% in Berkshire Hathaway. From 2020 the rates are 15.0%, 10.9% and 15.2%; from 2023 11.4%, 12.0% and 14.2%; from 2025, over less than two years, 11.3%, 10.5% and 11.1%. The wallet's value swings more than either comparison, because five companies of a few industries are far less diversified than the index or Berkshire.

The value wallet with sells buys the same picks with the same money, but at each decision it also sells part of any holding whose score has fallen; Section 4 describes the rule. From 2016 it is worth 4,623 USD (12.8% a year), from 2020 2,383 USD (14.8%) and from 2025 450 USD (11.4%).



(a) From 2016

(b) From 2020

(c) From 2023

(d) From 2025

Figure 2: Value of 100 USD every January and July from each start year, in USD, next to the money put in; the S&P MidCap 400 dashed and Berkshire Hathaway dash-dot.

4 Selling below the threshold

Table 3: Sales of the investor who started in 2020: share of each holding sold, its score on the 0-to-1 scale, and the proceeds.

Year	Sold: share of the holding (score)	Proceeds, USD
2021-07	GAP 10% (score 0.36)	3.95
2023-07	MMS 9% (score 0.36)	2.15
2024-01	GAP 4% (score 0.39)	0.92
2024-07	MKSI 39% (score 0.25), MMS 5% (score 0.38)	32.99
2025-01	MKSI 43% (score 0.23)	18.19
2025-07	MKSI 31% (score 0.28)	7.04
2026-01	MKSI 34% (score 0.27)	8.83
2026-07	MKSI 36% (score 0.26), WLK 2% (score 0.39)	16.27

Each January and July, before buying, the wallet with sells checks every company it holds. The score is put on a scale from 0 to 1, the share of the most the company could score on the rules it is measured on. A holding scoring 0.4 or less is sold in part: the share sold is (0.4 less the score) divided by 0.4, so all of it at a score of 0 and almost none just under 0.4. Holdings are sold, never sold short. The proceeds are added to the decision's 100 USD and buy its picks. Table 3 lists the 10 sales of the investor who started in 2020, worth 90 USD in all.

5 Wallet contents

Figure 3 follows every company in the wallet with sells of the investor who started in 2020, month by month. A line starts or steps up when the company is bought, steps down when part of it is sold, and rises or falls with the price in between. The dashed lines follow the same money in the S&P MidCap 400 and in Berkshire Hathaway, divided by the number of companies the wallet holds on the date, so they show what one holding would be worth in the index or in Berkshire. Appendix B gives each holding's value after every January's trades, the money put in and the proceeds of sales.

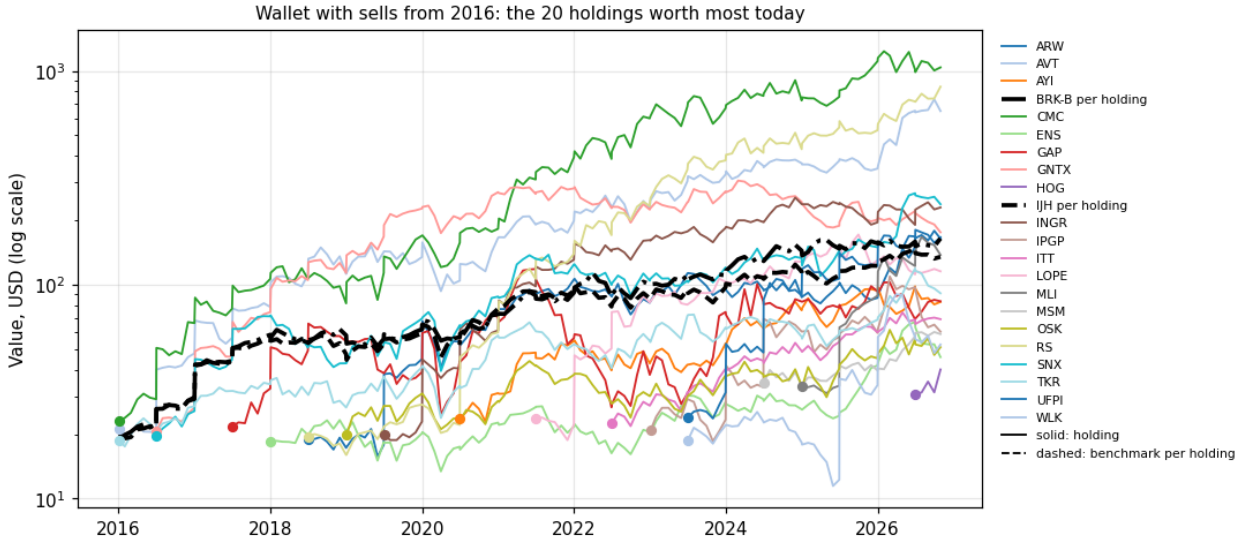


Figure 3: Value of each holding in the wallet with sells (solid) and of the S&P MidCap 400 and Berkshire Hathaway divided by the number of holdings (dashed), in USD on a logarithmic scale.

6 Rules behind the winners

Table 4: Dollars ahead of the same money in the S&P MidCap 400, per pick: total, average and median of the picks that beat the index and of those that lagged it.

Picks	Number	USD ahead, total	Average	Median
Beat the S&P MidCap 400	60	+1,109	+18.5	+12.1
Lagged the S&P MidCap 400	50	-593	-11.9	-10.1
All picks	110	+516	+4.7	+2.3

Of the picks, 60 have done better than the S&P MidCap 400 since their purchase and 50 have not, but a count treats a large win and a small loss alike. Table 4 weighs them in dollars: each pick's value today less what the same money would be worth in the S&P MidCap 400. The winners are 1,109 USD ahead of the index and the laggards 593 USD behind, a net 516 USD, the wallet's whole lead over the index. The median winner is 12 USD ahead and the median laggard 10 USD behind; the average win is 1.6 times the average loss. The wins are concentrated: the 5 largest, RS Jul 2018, CMC Jan 2016, RS Jul 2020, SNX Jul 2016 and RS Jan 2021, make up 27% of the winners' total.

Figure 4 compares the mean points per rule of the winners and of the laggards on the day they were picked, each pick weighted by its dollars ahead or behind the index, so the large wins and losses count most; the rules are ordered by how much more the winners scored. The winners scored most above the others on net income growth (NI1), 20 years of dividends (G4), P/E times P/B (G7) and price to tangible assets (E6); the picks that lagged scored higher on return on equity (B1) and 10 years of profit (G3). With so few picks, these differences describe this sample; they do not establish

which rules work. Appendix E explains each rule, and Appendix C tests whether the rule weights favour the rules that told winners from losers.

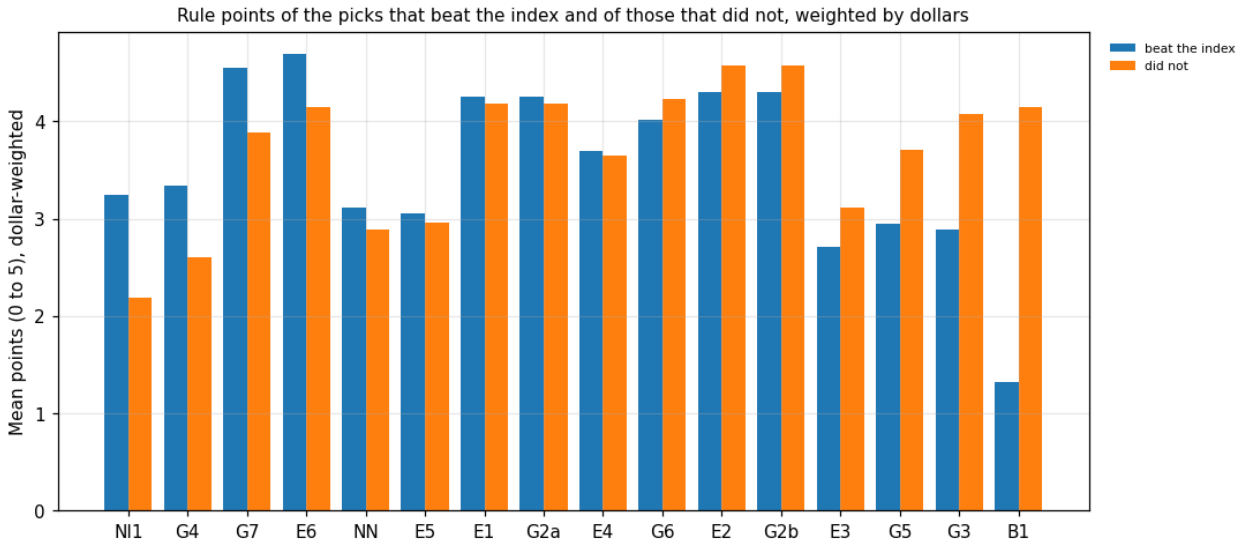


Figure 4: Mean points per rule (0 to 5) of the picks that beat the S&P MidCap 400 since purchase and of those that did not, each pick weighted by its dollars ahead or behind the index.

7 Price history of the picks

Figure 5 shows the share price of every company the investor who started in 2023 bought or sold, and of the next picks, over the last 5 years. The price is split-adjusted and excludes dividends, as a price chart shows it; the companies run from the best to the worst price change over the 5 years, given beside each name; when they do not fit on one page, only the 5 best and the 5 worst are shown. Each purchase is marked in green and each sale in red, at the price on the decision date; a larger marker means more money. Figure 6 gives the same trades in USD per company and decision date.

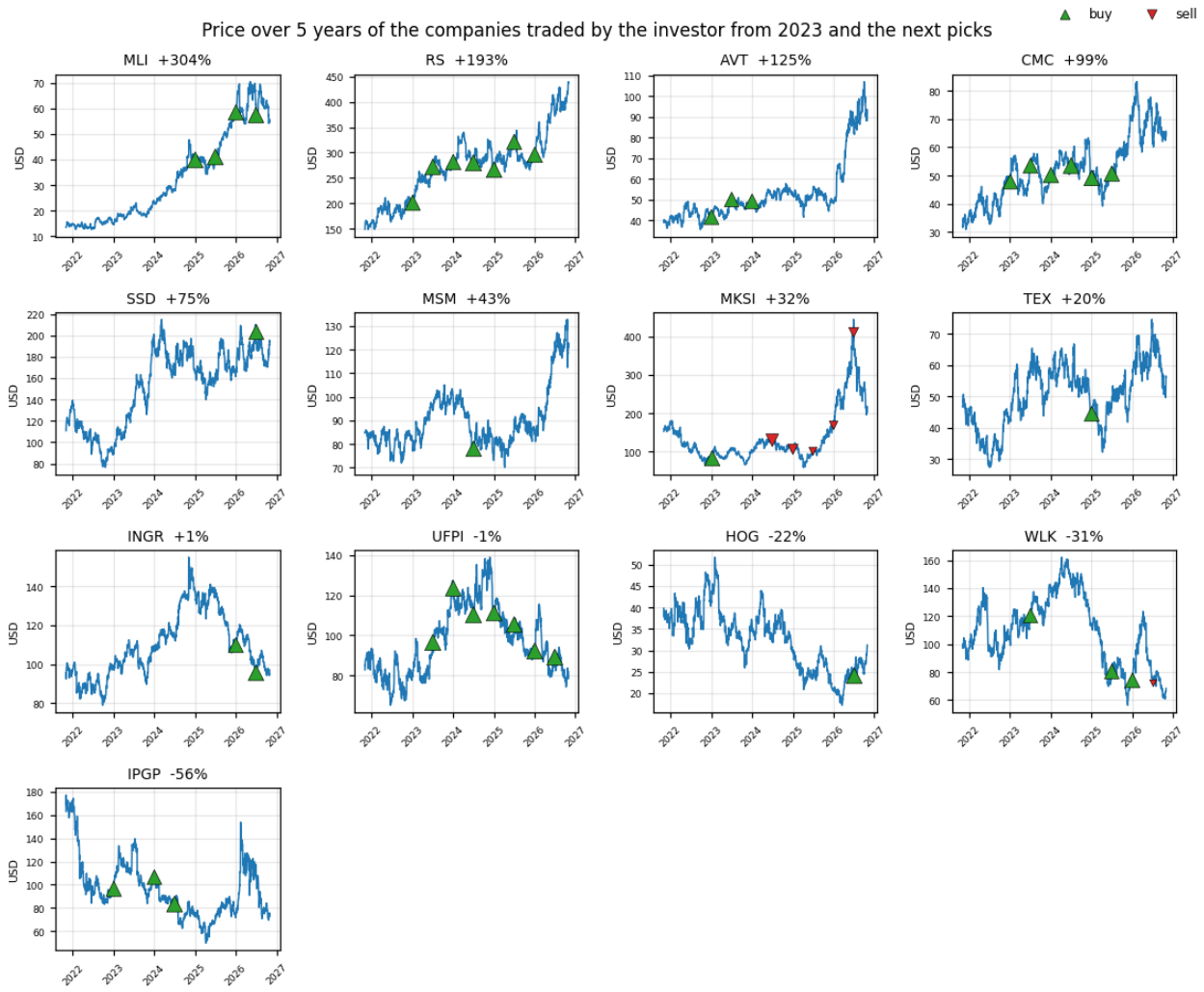


Figure 5: Share price (USD, split-adjusted, dividends excluded) over the last 5 years, one panel per company, ordered from the best to the worst price change over the period; purchases (green) and sales (red) of the investor who started in 2023, marker area in proportion to the USD traded.

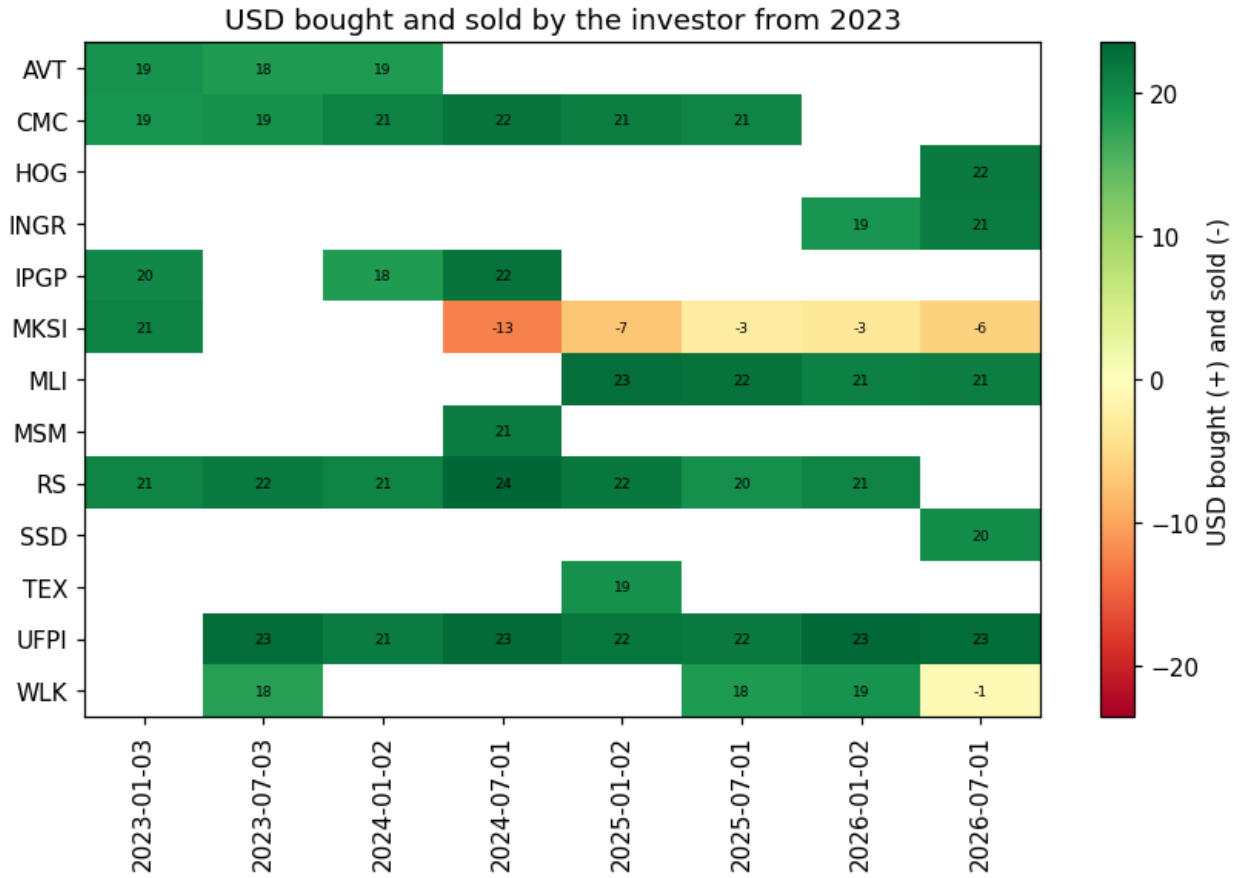


Figure 6: USD bought (green, positive) and sold (red, negative) by the investor who started in 2023, per company and decision date.

8 The next picks

Table 5: The next picks on the latest score date: score, score on the 0-to-1 scale, weight and USD of each 100 USD.

Company	Sector	Score	Score 0 to 1	Weight	USD
UFP Industries (UFPI)	Industrials	41.8	0.84	22.5%	22.53
Ingredion (INGR)	Staples	39.0	0.78	21.0%	21.04
Simpson Manufacturing (SSD)	Industrials	36.2	0.73	19.5%	19.51
Harley-Davidson (HOG)	Discretionary	34.4	0.77	18.5%	18.52
Reliance, Inc. (RS)	Materials	34.1	0.69	18.4%	18.40

On the latest score date, 18 September 2026, the five highest-scoring companies are UFP Industries (UFPI), Ingredion (INGR), Simpson Manufacturing (SSD), Harley-Davidson (HOG) and Reliance, Inc. (RS) (Table 5). Each would receive its share of the next decision's 100 USD and of any proceeds from sales. The decision is taken on the first trading day of January with the scores known then, which include the annual reports filed until that day; these picks are a preview and may change. Figure 7 shows each pick's share price over the last 5 years, with the earlier purchases of the investor who started in 2023 and the potential purchase at today's price.



Figure 7: Share price of each next pick (USD, split-adjusted, dividends excluded) over the last 5 years: earlier purchases of the investor who started in 2023 in green, the potential purchase at the latest price in orange.

8.1 Sell check

Table 6: Holdings the rule would sell in part: value today, score on the 0-to-1 scale, share to sell and amount.

Company	Holding, USD	Score 0 to 1	Sell	Sell, USD
Westlake Corporation (WLK)	46.14	0.39	1%	0.62
MKS Instruments (MKSI)	13.31	0.26	36%	4.81

The investor who started in 2020 holds 21 companies. With the scores of 18 September 2026, the rule would sell part of Westlake Corporation (WLK), 1% and MKS Instruments (MKSI), 36%, and keep the other 19 as they are (Table 6).

9 Conclusions

In this sample, the value picks bought twice a year and held are worth more today than the same money in the S&P MidCap 400 or in Berkshire Hathaway for each of the four start years. That describes the past under rules and settings chosen while looking at it; it is not a test of the strategy. The lead is uneven: the 12 purchases of Reliance, Inc. (RS) make up 25% of the 2020 investor's value, and the purchases of January 2018, January 2019, January 2020, July 2023, January 2024, July 2024, July 2025 and January 2026 trail the index.

Before the strategy is used, three steps are needed: the companies that left the index must be added to the data, the score must be tested on more decisions (each quarter instead of each year), and trading costs and taxes must be included.

A Every purchase

Table 7: Every purchase of the wallet with sells of the investor who started in 2020: weight in the decision's purchase, USD invested (the decision's 100 USD and the proceeds of sales), split-adjusted price at purchase and today, USD sold from it later, USD it keeps today and its total return with dividends, (sold + kept) / invested - 1.

Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2020-01	AVT	Technology	20%	20	42.72	88.20		50	+144%
2020-01	ARW	Technology	20%	20	85.79	208.34		49	+143%
2020-01	MKSI	Technology	20%	20	111.83	202.04	24	4	+37%
2020-01	INGR	Staples	20%	20	92.95	96.97		25	+27%
2020-01	GAP	Discretionary	20%	20	17.28	22.32	5	27	+65%
2020-07	RS	Materials	21%	21	92.97	436.91		109	+421%
2020-07	AVT	Technology	21%	21	26.90	88.20		80	+282%
2020-07	AYI	Industrials	20%	20	92.06	318.14		72	+252%
2020-07	INGR	Staples	19%	19	81.14	96.97		28	+43%
2020-07	CMC	Materials	18%	18	19.87	63.98		65	+255%
2021-01	RS	Materials	22%	22	119.71	436.91		86	+300%
2021-01	INGR	Staples	21%	21	78.86	96.97		30	+45%
2021-01	CMC	Materials	20%	20	20.45	63.98		68	+240%
2021-01	MMS	Industrials	19%	19	73.07	63.72	3	16	-1%
2021-01	LSTR	Industrials	18%	18	133.09	142.70		22	+21%
2021-07	LOPE	Discretionary	21%	22	90.88	147.88		36	+63%
2021-07	INGR	Staples	20%	21	90.99	96.97		26	+24%
2021-07	RS	Materials	20%	21	152.92	436.91		65	+211%
2021-07	CMC	Materials	20%	20	31.14	63.98		45	+121%
2021-07	LPX	Industrials	19%	20	60.50	73.31		26	+30%
2022-01	CMC	Materials	22%	22	36.34	63.98		42	+88%
2022-01	LOPE	Discretionary	20%	20	87.47	147.88		35	+69%
2022-01	INGR	Staples	20%	20	97.12	96.97		23	+14%
2022-01	RS	Materials	19%	19	161.28	436.91		54	+192%
2022-01	AVT	Technology	18%	18	41.10	88.20		44	+141%
2022-07	RS	Materials	21%	21	171.07	436.91		57	+173%
2022-07	LOPE	Discretionary	21%	21	95.37	147.88		32	+55%
2022-07	ITT	Industrials	20%	20	67.50	198.50		60	+207%
2022-07	CMC	Materials	20%	20	33.81	63.98		39	+100%
2022-07	MKSI	Technology	19%	19	96.32	202.04	26	4	+56%
2023-01	MKSI	Technology	21%	21	84.67	202.04	32	5	+77%
2023-01	RS	Materials	21%	21	201.39	436.91		48	+130%
2023-01	IPGP	Technology	20%	20	96.49	72.59		15	-25%
2023-01	AVT	Technology	19%	19	41.59	88.20		44	+132%
2023-01	CMC	Materials	19%	19	47.83	63.98		27	+41%
2023-07	UFPI	Industrials	23%	23	96.50	81.92		21	-11%
2023-07	RS	Materials	22%	22	272.16	436.91		37	+69%
2023-07	CMC	Materials	19%	20	53.47	63.98		25	+25%
2023-07	AVT	Technology	18%	19	50.09	88.20		36	+90%
2023-07	WLK	Materials	18%	18	120.38	68.28	0	11	-39%
2024-01	UFPI	Industrials	21%	22	123.56	81.92		15	-31%
2024-01	CMC	Materials	21%	21	50.18	63.98		28	+33%
2024-01	RS	Materials	21%	21	281.43	436.91		34	+62%
2024-01	AVT	Technology	19%	19	49.18	88.20		36	+91%
2024-01	IPGP	Technology	18%	19	106.58	72.59		13	-32%
2024-07	RS	Materials	21%	28	279.98	436.91		45	+62%
2024-07	UFPI	Industrials	21%	27	110.32	81.92		21	-23%
2024-07	IPGP	Technology	20%	26	83.33	72.59		23	-13%
2024-07	CMC	Materials	20%	26	53.52	63.98		32	+23%
2024-07	MSM	Industrials	19%	25	78.23	121.02		43	+69%
2025-01	MLI	Industrials	21%	25	39.87	54.56		35	+40%

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Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2025-01	UFPI	Industrials	20%	24	111.16	81.92		18	-24%
2025-01	RS	Materials	20%	24	266.61	436.91		40	+68%
2025-01	CMC	Materials	20%	24	49.15	63.98		31	+34%
2025-01	TEX	Industrials	18%	21	44.52	56.15		28	+29%
2025-07	MLI	Industrials	22%	23	41.03	54.56		31	+35%
2025-07	UFPI	Industrials	21%	22	105.47	81.92		18	-21%
2025-07	CMC	Materials	20%	22	50.71	63.98		28	+28%
2025-07	RS	Materials	19%	20	321.33	436.91		28	+39%
2025-07	WLK	Materials	18%	19	80.48	68.28	0	17	-12%
2026-01	UFPI	Industrials	22%	24	92.18	81.92		22	-10%
2026-01	MLI	Industrials	20%	22	58.40	54.56		21	-6%
2026-01	RS	Materials	20%	22	295.82	436.91		33	+49%
2026-01	WLK	Materials	19%	20	74.15	68.28	0	19	-6%
2026-01	INGR	Staples	18%	20	109.79	96.97		18	-9%
2026-07	UFPI	Industrials	21%	25	89.15	81.92		23	-8%
2026-07	HOG	Discretionary	20%	23	24.15	31.22		31	+30%
2026-07	INGR	Staples	20%	23	95.62	96.97		24	+2%
2026-07	MLI	Industrials	20%	23	57.42	54.56		22	-5%
2026-07	SSD	Industrials	19%	22	203.42	193.34		21	-5%

B Wallet contents by year

Table 8: Wallet with sells of the investor who started in 2020: value of each holding after each January's trades and today (USD), and the money put in and the proceeds of sales to date, July decisions included.

Holding	2020	2021	2022	2023	2024	2025	2026	Now
ARW	20	23	32	25	28	26	27	49
AVT	20	45	72	94	151	163	160	290
GAP	20	22	19	13	24	27	30	27
INGR	20	57	116	120	139	177	166	174
MKSI	20	27	32	54	64	24	17	13
AYI		27	47	38	45	67	84	72
CMC		39	117	203	255	301	476	429
RS		49	108	182	303	342	426	636
MMS		19	22	20	21	18	21	16
LSTR		18	25	23	28	25	22	22
LPX			26	20	24	36	28	26
LOPE			41	74	91	112	114	102
ITT				24	35	42	53	60
IPGP				20	41	51	53	51
UFPI					52	99	127	138
WLK					22	17	50	46
MSM						25	29	43
MLI						25	93	109
TEX						21	27	28
HOG								31
SSD								21
Put in	100	300	500	700	900	1,100	1,300	1,400
Sold	0	0	4	4	7	58	74	90
Total	100	327	657	909	1,322	1,599	2,005	2,383

C Testing the weighting

Table 9: Each rule's weight w_r on the picks' dates (overall, for the picks that beat the S&P MidCap 400 and for the others), the dollar-weighted mean points, the gap and the rank correlation with the return a year above the S&P MidCap 400.

Rule	w_r	w_r beat	w_r not	Pts beat	Pts not	Gap	Corr.
NN	1.00	1.00	1.00	3.12	2.88	+0.23	+0.00
E6	0.99	0.99	0.99	4.69	4.15	+0.54	+0.10
G7	0.84	0.84	0.84	4.55	3.88	+0.67	+0.26
G6	0.82	0.82	0.82	4.01	4.23	-0.22	+0.07
B1	0.79	0.81	0.77	1.32	4.14	-2.82	-0.38
G2b	0.69	0.69	0.70	4.30	4.57	-0.27	-0.04
E2	0.66	0.66	0.66	4.30	4.57	-0.27	-0.04
G2a	0.61	0.61	0.61	4.26	4.18	+0.08	-0.01
G4	0.49	0.49	0.48	3.34	2.60	+0.73	+0.06
G3	0.40	0.41	0.39	2.88	4.08	-1.19	-0.29
NI1	0.40	0.39	0.40	3.25	2.18	+1.07	+0.21
E1	0.37	0.37	0.37	4.26	4.18	+0.08	-0.01
E4	0.32	0.32	0.33	3.69	3.65	+0.05	+0.14
E3	0.29	0.29	0.29	2.71	3.11	-0.40	-0.14
E5	0.26	0.26	0.26	3.05	2.96	+0.09	+0.15
G5	0.20	0.21	0.19	2.95	3.71	-0.76	-0.28

Each rule's points are multiplied by its weight, one less the share of companies passing it on the date, so a rule few companies pass counts more. A good weighting gives most weight to the rules whose points went with better returns. Table 9 sets each rule's mean weight on the dates the picks were made against two measures of how well it told the 110 picks apart: the gap between the mean points of the picks that beat the S&P MidCap 400 and of those that did not, weighted by dollars as in Figure 4, and the rank correlation of the rule's points with each pick's return a year above the S&P MidCap 400. Figure 8 plots weight against gap; a good weighting puts its rules on a rising line, with the heavy rules above zero.

Across the rules, the rank correlation of weight and gap is +0.23, and of weight and correlation +0.14.

The weighting is unrelated to which rules worked. Weighted heavily but with a negative gap: debt to working capital (G2b), P/E (G6), debt to working capital (lenient) (E2) and return on equity (B1). A positive gap but a light weight: 20 years of dividends (G4), current ratio (lenient) (E1), current dividend (E4), 4-year earnings growth (E5) and net income growth (NI1). Any change to the weights must be tested on years it was not fitted on.

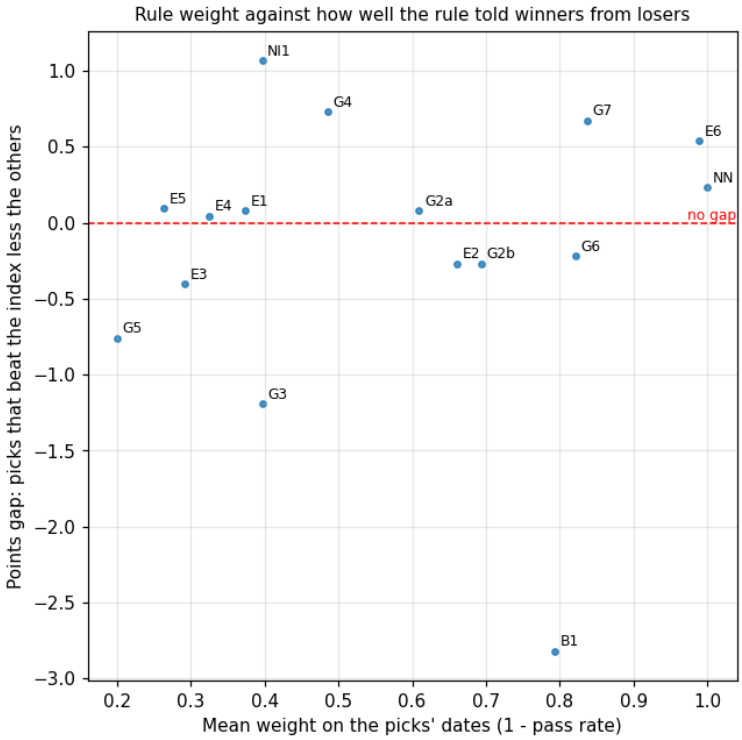


Figure 8: Each rule's mean weight against the gap in points between the picks that beat the S&P MidCap 400 and the others, weighted by dollars; rules above the line told winners from losers.

D Forward information

Table 10: Truncation check: companies scored on the date, companies whose score differs when every input is cut at the date (per score), and whether the five picks are the same.

Date	Companies	Score	Score 0 to 1	Unweighted	Share passed	Same picks
2018-12-21	128	0	0	0	0	yes
2022-12-16	187	0	0	0	0	yes
2025-12-19	272	0	0	0	0	yes

A score must use only what was known on its date. Four guards are built in. A company's figures count from the day its annual report was filed with the SEC, and of all the figures filed for a year only the first reported is used, never a later restatement. Prices are the close on the scoring date and dividends are counted by their ex-date. Each January and July purchase uses the scores of the rebalance date two weeks before it. Rule weights come from the companies on the date alone.

These guards are tested by cutting every input at a date as if it were that day (filings filed on or before it, prices and dividends up to it), scoring the companies again and comparing the scores with the full run (Table 10). Any difference means information from after the date reached a score. On 21 December 2018, 16 December 2022 and 19 December 2025 the picks were the same, and the scores were identical on 21 December 2018, 16 December 2022 and 19 December 2025. The same check runs as a unit test. A rule whose window reaches back further than a company's price history (the 20-year dividend record of a younger company) is not evaluated rather than failed.

Three sources of forward information are outside this test. Only today's S&P MidCap 400 members are scored. The index's list gives no entry dates: a company counts from its latest addition in the list of changes, or from the start of that list (January 2012) when no addition is listed. The list holds selected changes only, so some companies count before they joined: a look-ahead that flatters the results. Companies that later left the index are missing, which flatters every result. Sectors are

today's, used to leave out Financials and Real Estate in earlier years. And the score's design and settings were chosen while looking at 2016 to 2026, so no period tests it fairly: 2022 onwards, once meant as a held-out test, was used to choose between models.

D.1 Companies missing from the data

Table 11: Companies that left the S&P MidCap 400 since each start year, all missing from the data: in all, on their market value, and through an acquisition or another reason.

Start	Left the index	On market value	Acquired or other
2016	468	217	251
2020	293	203	90
2023	164	123	41
2025	65	41	24

Table 11 counts the companies that left the S&P MidCap 400 after each start year; none of them is in the data, of any sector. Most left through an acquisition, often at a premium; those that dropped out on their market value are the likely losers a real investor could have picked. The later the start, the fewer are missing, which is why the investor who started in 2020 is the reference of this report. The count starts with the index's list of changes, which begins in January 2012: removals before that are not counted, so for earlier start years it understates the gap. The bias is removed only when the historical members of the index are added to the data.

E The rules

Table 12: The rules: what each measures, the better direction and when a company passes.

Rule	Name	What it measures	Better	Passes when
G2a	current ratio	Current assets divided by current liabilities: what the company owns against what it owes within a year.	higher	at least 2.0
G2b	debt to working capital	Long-term debt divided by net current assets (current assets less current liabilities).	lower	at most 1.0
G3	10 years of profit	Number of the last 10 fiscal years with positive earnings per share.	higher	all 10 years
G4	20 years of dividends	Number of the 20 calendar years before the decision in which a dividend was paid.	higher	all 20 years
G5	10-year earnings growth	Average earnings per share of the last 3 years over the average of 3 years a decade earlier.	higher	at least 1.33
G6	P/E	Price over the average earnings per share of the last 3 years.	lower	at most 15.0
G7	P/E times P/B	P/E as in G6 times the price over book value per share.	lower	P/B at most 1.5, or the product at most 22.5
E1	current ratio (lenient)	Current ratio, as G2a.	higher	at least 1.5
E2	debt to working capital (lenient)	Long-term debt over net current assets, as G2b.	lower	at most 1.1
E3	5 years of profit	Number of the last 5 fiscal years with positive earnings per share.	higher	all 5 years
E4	current dividend	Dividends paid over the last 365 days, as a yield on the price.	higher	any dividend
E5	4-year earnings growth	Change in earnings per share over 4 years, relative to the earlier figure.	higher	earnings per share grew
E6	price to tangible assets	Market value over net tangible assets (equity less goodwill and other intangible assets).	lower	at most 1.2
NN	net-net	Market value over net current asset value (current assets less all liabilities).	lower	at most 0.67
B1	return on equity	Number of the last 10 years with return on equity (net income over equity) of at least 15%.	higher	at least 7 years, and long-term debt at most 0.5 times equity
NI1	net income growth	Growth of net income from the year before to the latest year.	higher	net income grew

Table 12 explains each rule: what it measures, whether a higher or a lower value earns more points, and when a company passes it. The points come from the company's rank on the measure among all companies on the date; the pass threshold sets the rule's weight. Rules G follow Graham's defensive investor (The Intelligent Investor, chapter 14), rules E his enterprising investor (chapter 15) and his net-net test (NN), B1 Buffett's return on equity, and NI1 was added for net income growth. Two pairs test the same measure at different strictness (G2a and E1, G2b and E2), so the current ratio and debt count twice.